

# Research Methodology in CS (CSEG3060\_3)

## Lecture 3: Licensing of Patents - types of licenses (exclusive, non-exclusive, sole), licensing agreements, and royalty structures

Unit 6: Patent Rights and New Developments in IPR • Mapped Course Outcome: CO3

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### Lecture 3: Licensing of Patents — Types of Licenses, Licensing Agreements, and Royalty Structures

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**Course:** Research Methodology in Computer Science (CSEG3060\_3) **Program:** B.Tech (Computer Science) **Semester:** 5 **Unit:** Unit 6 — Patent Rights and New Developments in IPR **Lecture:** 3 of the Unit **Mapped Course Outcome:** CO3 — Demonstrate an understanding of intellectual property rights with the ability to make informed judgments on IPR-related issues in research and professional practice. **Estimated Reading Time:** 50–60 minutes

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#### 1. Learning Objectives

By the end of this lecture, students should be able to:

1. Define patent licensing and distinguish it from patent assignment.
  2. Identify and compare the principal types of patent licenses — exclusive, non-exclusive, and sole.
  3. Identify and explain the principal clauses of a patent licensing agreement.
  4. Analyze the structure and computation of royalties, including lump-sum, running, percentage, and milestone-based models.
  5. Evaluate licensing arrangements using case studies drawn from the information and communication technology (ICT) sector.
  6. Apply licensing principles to academic and entrepreneurial contexts relevant to computer science research.
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#### 2. Introduction

A patent grants its owner a time-limited, territorially bounded monopoly to exclude others from making, using, offering for sale, selling, or importing the patented invention. However, ownership

of a patent does not automatically translate into commercial exploitation, technological diffusion, or revenue realization. Many patent owners — including individual inventors, universities, and even large technology firms — lack the manufacturing capacity, distribution networks, or technical infrastructure required to commercialize the invention themselves. In such situations, **licensing** becomes the central legal-commercial instrument that enables the controlled transfer of patent rights from one party to another without relinquishing ownership.

Licensing is the commercial and contractual backbone of the modern technology economy. It underpins standards such as Wi-Fi, 4G/5G, MPEG, JPEG, and USB, and it shapes the revenue models of firms whose principal assets are intangible intellectual property. For computer science researchers, understanding patent licensing is essential because:

- Most enterprise software revenue is generated through licensing models.
- Open-source and proprietary software coexist primarily because of well-defined licensing frameworks.
- Researchers working in industrial or university Technology Transfer Offices (TTOs) regularly encounter licensing agreements.
- Joint research, sponsored research, and consulting arrangements frequently include embedded patent licensing clauses.

This lecture examines the three principal dimensions of patent licensing: (a) the typology of licenses, (b) the structure of the licensing agreement, and (c) the royalty frameworks through which licensing generates economic value.

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### 3. Key Concepts

| Term                       | Working Definition                                                                                                                                                                     |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Patent</b>              | A statutory right granted by the State conferring on the patentee the exclusive right to exclude others from exploiting the invention for a limited period.                            |
| <b>Licensing</b>           | A contractual arrangement whereby the patent owner (licensor) authorizes another party (licensee) to perform one or more acts covered by the patent, subject to defined conditions.    |
| <b>Licensor</b>            | The party that owns or controls the patent and grants the license.                                                                                                                     |
| <b>Licensee</b>            | The party that receives the license and exercises the licensed rights.                                                                                                                 |
| <b>Licensing Agreement</b> | The written contract that sets out the rights, obligations, royalties, term, territory, field of use, termination conditions, and dispute resolution mechanisms governing the license. |
| <b>Royalty</b>             | The consideration paid by the licensee to the licensor for the use of the licensed patent rights, typically calculated on a per-unit, percentage, lump-sum, or milestone basis.        |
| <b>Technology Transfer</b> | The broader process by which technical know-how, patents, trademarks, and tacit knowledge move from an originator to a recipient, of which licensing is one mechanism.                 |
| <b>Patent Pool</b>         | An aggregation of patents from multiple owners, licensed collectively to participants and/or third parties, often to support industry standards.                                       |
| <b>Cross-Licensing</b>     | A bilateral or multilateral arrangement in which two or more parties license their respective patents to one another.                                                                  |

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### 4. Detailed Explanation

#### 4.1 Patent Licensing: Conceptual Foundations

Patent licensing is the contractual act of granting permission to use a patented invention. Legally, a license is a personal privilege — it does not transfer ownership of the patent. The patentee retains title to the patent, and the licensee obtains only those rights expressly enumerated in the agreement. This is a fundamental distinction between licensing and assignment (also known as assignment of title), which is a complete transfer of ownership.

The legal basis for patent licensing is found in patent statutes such as the Indian Patents Act, 1970 (Section 68 onwards), the U.S. Patent Act (35 U.S.C. § 261), and the European Patent Convention (Article 73). Under these regimes, patents are treated as movable property and may be licensed, assigned, or mortgaged.

Patent licensing serves multiple economic and strategic objectives:

- **Revenue generation** through royalties and fees.
- **Market expansion** by leveraging the licensee's established distribution channels.
- **Risk sharing**, particularly where commercialization requires substantial capital investment.
- **Standardization** by enabling coordinated use of patents essential to industry standards (e.g., 5G, Wi-Fi).
- **Defensive monetization** to deter litigation.
- **Technology diffusion** and societal benefit, particularly in university and public-sector contexts.

In the computer science domain, licensing has acquired a dominant position because software-related inventions, communication protocols, and hardware designs are typically patented and rarely commercialized through in-house production alone.

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## 4.2 Types of Patent Licenses

Patent licenses are categorized primarily along three axes: (a) exclusivity, (b) scope (territorial and field-of-use), and (c) duration. The most commercially significant dimension is exclusivity, which produces the canonical trilogy of license types: **exclusive**, **non-exclusive**, and **sole**.

### 4.2.1 Exclusive License

An exclusive license grants the licensee the sole right to practice the licensed patent within a defined territory and field of use, to the exclusion of all other parties — including the licensor. While the patentee retains ownership, the licensor is contractually barred from exercising those same rights and from granting further licenses.

#### Key Characteristics:

- Only one licensee is granted the right.
- The licensor itself is excluded from commercial use within the agreed scope.
- The licensee is often required to meet performance milestones (e.g., minimum production or sales volumes) to maintain exclusivity.
- The licensee usually has the right (subject to agreement) to sue infringers in its own name.

- Most commercially valuable to the licensee and therefore most expensive for the licensee to obtain.

**Typical Use Cases:**

- University spin-out commercialization.
- Pharmaceutical licensing of blockbuster molecules (where one licensee manufactures globally).
- Semiconductor Foundries licensing process patents from a fabless design house.

**Considerations for Computer Science:**

- Exclusive licensing is common in specialized embedded systems, AI accelerators, and proprietary database engines.
- Risks include "shelving" of inventions if the exclusive licensee lacks incentive to commercialize broadly.

**4.2.2 Non-Exclusive License**

A non-exclusive license permits the licensee to use the patent while the licensor retains the right to grant similar licenses to other parties and to use the patent itself. There is no restriction on the number of licensees.

**Key Characteristics:**

- Any number of licensees can hold parallel rights.
- The licensor continues to practice the invention.
- Royalty rates are typically lower because the exclusivity premium is absent.
- Used where the patent covers an industry standard or where the licensor seeks maximum diffusion.

**Typical Use Cases:**

- Standard-essential patents (SEPs) licensed through patent pools (e.g., MPEG-LA, Sisvel).
- Codecs, image formats, communication protocols (H.264/AVC, H.265/HEVC, 4G LTE).
- Open specifications where multiple implementations must coexist.

**Considerations for Computer Science:**

- Non-exclusive licensing is the dominant model for SEPs that enable interoperability.
- The "Fair, Reasonable, and Non-Discriminatory" (FRAND) commitment is typically associated with non-exclusive licensing in standards contexts.

### 4.2.3 Sole License

A sole license is a hybrid between the exclusive and non-exclusive forms. The licensor grants the licensee the right to practice the patent, retains its own right to practice, but undertakes not to grant additional licenses to any third party.

#### Key Characteristics:

- Only two parties may practice the patent: the licensor and the sole licensee.
- The licensor cannot issue further licenses.
- The licensee does not enjoy the right to exclude the licensor.

#### Typical Use Cases:

- Joint research arrangements where one party retains background rights and grants a sole license for foreground developments.
- Manufacturing arrangements where the patent owner also produces competing goods.

#### Comparative Summary:

| Dimension                       | Exclusive | Sole                  | Non-Exclusive |
|---------------------------------|-----------|-----------------------|---------------|
| Number of licensees             | One       | One                   | Unlimited     |
| Licensor may practice           | No        | Yes                   | Yes           |
| Licensor may sublicense further | No        | No (to third parties) | Yes           |
| Royalty magnitude               | Highest   | Intermediate          | Lowest        |
| Risk to licensor                | Highest   | Intermediate          | Lowest        |

### 4.2.4 Related Structures

For completeness, four additional structures warrant brief recognition:

- **Cross-License:** Two or more parties exchange licenses to their respective patents. Cross-licensing is widespread in high-tech where firms hold dense, overlapping patent portfolios (e.g., smartphone "peace" agreements).
- **Compulsory License:** Imposed by a State authority on the patentee in defined public-interest circumstances — for instance, where the patented invention is not worked in the territory, where a patented invention relates to semiconductor technology under specific national laws, or where a pharmaceutical patent is the subject of a government use notification. The patented

vaccine landscape during the COVID-19 pandemic brought these powers to international attention.

- **Patent Pool License:** A single license that bundles multiple patents owned by different entities. Pools are typical for standards-implementing technologies.
  - **Sublicense:** A license granted by an existing licensee to a third party. Sublicensing typically requires explicit authorization in the primary license.
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### 4.3 Licensing Agreements

A licensing agreement is the binding legal document that operationalizes a license. While the structure is shaped by negotiation, statutory norms, and industry custom, mature licensing agreements share a robust anatomy of clauses. The following sections systematically examine each component.

#### 4.3.1 Recitals and Definitions

The opening section establishes the parties, the recitals (the background and intent), and a definitions subsection. Definitions should be unambiguous and exhaustive — particularly for terms such as "Licensed Product," "Licensed Process," "Territory," "Field of Use," "Net Sales," "Affiliate," and "Effective Date."

#### 4.3.2 Grant Clause

The grant clause is the operative provision. It must specify:

- Which patents are licensed (by number, country, and family).
- Whether the license is exclusive, sole, or non-exclusive.
- Whether sublicensing rights are granted, and if so, under what conditions.
- The field of use (e.g., "use in consumer electronics but not in automotive applications").
- The territory (e.g., "India and Bhutan" or "worldwide").

**Drafting Note:** The grant clause is the most heavily litigated provision. Ambiguity in the scope of the grant almost always favours the licensee under customary contract interpretation principles.

#### 4.3.3 Consideration and Royalty Provisions

This clause specifies the financial consideration. It is usually framed by reference to one of the royalty structures discussed in Section 4.4.

#### 4.3.4 Performance Obligations

Particularly in exclusive or sole licenses, the licensor protects itself by requiring the licensee to:

- Achieve minimum sales or production volumes by specified dates.
- Use commercially reasonable efforts to commercialize the licensed product.
- Pay minimum annual royalties.
- Report quarterly or annually on commercialization progress.

Failure to meet performance milestones typically triggers a reversion of rights.

#### 4.3.5 Improvements and Derivatives

This clause determines who owns improvements to the licensed invention:

- **Licensor ownership:** The licensee assigns improvements to the licensor.
- **Licensee ownership:** The licensee retains ownership but grants a license-back to the licensor.
- **Joint ownership:** Common in university–industry collaborations.

#### 4.3.6 Reporting, Auditing, and Record-Keeping

The licensee is required to submit periodic royalty reports showing units sold, net revenue, royalties due, and currency. The licensor typically retains audit rights (often annually) to verify the accuracy of these reports.

#### 4.3.7 Warranties and Indemnities

Standard warranties in a patent license include:

- Warranty of title (licensor owns the patent).
- Warranty of non-infringement of third-party rights (often contested).
- Warranty of validity (limited or disclaimed in many jurisdictions).

Indemnity clauses determine which party bears liability for third-party infringement claims.

#### 4.3.8 Term and Termination

The agreement specifies:

- The commencement date and the term (often tied to the patent's lifetime).
- Grounds for termination — breach, insolvency, material misrepresentation, change of control.

- Post-termination obligations, especially regarding inventory sell-off, transition support, and continuing royalty obligations.

#### 4.3.9 Confidentiality

Especially in technology licensing, the licensee typically gains access to confidential know-how. The confidentiality clause constrains disclosure to defined personnel and purposes.

#### 4.3.10 Governing Law and Dispute Resolution

The agreement selects the governing law and the dispute resolution mechanism (litigation, arbitration, mediation). Multi-jurisdictional licensing deals typically designate international arbitration (e.g., Singapore International Arbitration Centre, London Court of International Arbitration, or the International Chamber of Commerce).

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### 4.4 Royalty Structures

A **royalty** is the consideration paid by the licensee to the licensor in exchange for the licensed rights. The structure of the royalty determines the cash flow profile, the allocation of risk between the parties, and the strategic incentives embedded in the deal.

#### 4.4.1 Classification of Royalty Structures

Royalties can be grouped into four primary structures, often combined within a single agreement.

##### 1. Lump-Sum Payment

A single, fixed payment made at the outset or on the achievement of a milestone.

- *Advantages:* Certainty for the licensor; no ongoing accounting; simplicity.
- *Disadvantages:* Does not scale with licensee success; overpayment risk if the product fails.
- *Typical Uses:* University patent licensing, one-time software access fees, and early-stage technology transfer to established firms.

##### 2. Running Royalty (Percentage of Net Sales)

A periodic royalty calculated as a fixed percentage of the net sales price of the licensed product.

- *Advantages:* Scales with licensee revenue; aligns incentives.
- *Disadvantages:* Requires ongoing monitoring and audit.
- *Typical Range:* In software and ICT licensing, running royalty rates typically range from 1% to 12% of net sales, depending on the technology's contribution, exclusivity, and market position.

### 3. Per-Unit Royalty

A fixed monetary amount per unit sold (e.g., USD 0.50 per device, INR 1.00 per chip).

- *Advantages:* Simplicity; easy verification.
- *Disadvantages:* Does not capture value changes arising from price inflation or premium positioning.
- *Typical Uses:* Codec licensing (e.g., per-device MPEG-LA royalties), semiconductor IP licensing.

### 4. Milestone Payments

Payments linked to specific events such as regulatory approval, first commercial sale, achievement of sales thresholds, or completion of development stages.

- *Advantages:* Rewards development progress; offsets licensor's value over time.
- *Disadvantages:* Requires clear, objective milestones.
- *Typical Uses:* Pharmaceutical licensing; large-scale ICT development deals.

### 5. Minimum Annual Royalties

A floor below which the licensee must pay regardless of actual sales. Minimum royalties are characteristic of exclusive licensing agreements and serve to prevent under-utilization.

### 6. Combination Models

Most contemporary licensing agreements are hybrid: a small upfront lump sum, combined with milestone payments, combined with running royalties, combined with a minimum annual royalty.

#### 4.4.2 The Royalty Calculation

In its simplest form, a running royalty is computed as:

$$R = S \times r \quad RR = SS \times rr$$

where:

- $RR$  = Total royalty owed.
- $SS$  = Net sales of the licensed product over the relevant period.
- $rr$  = Royalty rate (expressed as a decimal, e.g., 0.05 for 5%).

When the licensed product incorporates multiple technologies or components, the royalty may be computed on the **smallest saleable patent-practicing unit (SSPPU)**, which prevents royalty

"stacking" and limits the licensor's claim to the contribution of its technology alone.

Where multiple licensees contribute patents to a single product, **royalty stacking** may occur if each licensor demands a percentage of the total product price, producing a combined royalty burden that exceeds commercially sustainable levels. Patent pools, aggregate royalty caps, and the SSPPU concept are all market responses to stacking.

### Worked Example 1: Running Royalty

A software firm licenses a patented data compression engine from a university. The license is non-exclusive, with a royalty rate of 4% of net sales. The licensee sells 250,000 units of the compression-enabled application at a net price of INR 800 per unit in Year 1.

$$R = 250,000 \times 800 \times 0.04 = 8,000,000$$

Thus, the royalty for the year is INR 8,000,000.

### Worked Example 2: Per-Unit Royalty with Minimum Floor

A chipset manufacturer licenses an error-correction technology at INR 1.20 per unit with a minimum annual royalty of INR 5,000,000. The licensee produces 3,500,000 units in the year.

Computed royalty:  $R = 3,500,000 \times 1.20 = 4,200,000$ . Minimum royalty: INR 5,000,000. Royalty due (whichever is higher): INR 5,000,000.

### Worked Example 3: Combined Lump-Sum and Running Royalty with Milestone

A robotics start-up licenses an obstacle-detection patent from a research institute:

- Upfront lump-sum: INR 4,000,000.
- Milestone payment on first commercial sale: INR 2,000,000.
- Running royalty: 3% of net sales.
- Minimum annual royalty after Year 2: INR 1,500,000.

In Year 1, the start-up records first commercial sale (milestone triggered), then sells 50,000 units at a net price of INR 25,000 each.

Milestone payment: INR 2,000,000. Running royalty:  $50,000 \times 25,000 \times 0.03 = 37,500,000$   
 $50,000 \times 25,000 \times 0.03 = 37,500,000$ . Total Year 1 royalty: INR 39,500,000 (assuming minimum annual does not yet apply).

### 4.4.3 Factors Influencing Royalty Rates

Royalty rates are negotiated outcomes, not statutory values. The principal determinants are:

- **Nature of the technology:** Foundational technologies commanding higher royalties.
- **Stage of development:** Early-stage, unproven technologies command lower rates.
- **Exclusivity:** Exclusive licenses command higher royalties.
- **Field of use:** Broad-field licenses command higher rates.
- **Market size and growth prospects.**
- **Available alternatives (design-arounds).**
- **Risk profile and development costs borne by the licensee.**
- **Standards-essentiality** (FRAND-constrained).
- **Strategic significance to the licensee** (e.g., blocking a competitor).
- **Comparable industry licensing benchmarks.**

#### 4.4.4 Tax Treatment of Royalties

Royalties are typically treated as "royalty income" in the country of receipt and are subject to withholding tax at source. International licensing may be affected by Double Taxation Avoidance Agreements (DTAAs). The classification of a payment as "royalty" or as "business income" materially affects taxation and is often contested in audits.

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#### 4.5 Licensing and Computer Software: Particular Considerations

The intersection of patent law and software is jurisdictionally uneven. Software-as-such is not patentable in many jurisdictions in the way it is in the United States (after the Supreme Court's *Alice Corp. v. CLS Bank International*, 573 U.S. 208 (2014) decision), although software implemented as part of a technical solution is patentable under Sections 3(k) and 10 of the Indian Patents Act, 1970 and Articles 52(2) and (3) of the European Patent Convention. Despite this unevenness, the dominant IP regime for software worldwide remains **copyright**, with patent protection layered on top for specific technical innovations.

Software licensing conventions include:

- **Proprietary / End-User License Agreements (EULAs).** Restrictive licensing to a single user or organization, often non-transferable.
- **OEM and Volume Licensing.** Bundled with hardware or distributed in bulk to enterprises.
- **SaaS Subscription Models.** Access-based rather than ownership-based.
- **Open-Source Licensing.** The GNU General Public License (GPL), MIT License, Apache License, and BSD License families establish conditions that directly affect patent rights (e.g., GPL's "patent grant" clause).

- **Patent Pools for Standardization.** Critical for codecs (MPEG-LA, Via Licensing), Wi-Fi, and mobile communications.

A deep understanding of licensing is therefore indispensable in contemporary CS practice, since the choice of license determines downstream obligations, including patent retaliation, defensive termination, and source-code disclosure.

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## 5. Case Studies and Practical Examples

### 5.1 Case Study A: MPEG-LA and the H.264 (AVC) Patent Pool

The MPEG-LA patent pool aggregates patents from multiple holders essential to the H.264 video compression standard. Licensees obtain a single license covering all pooled patents. The pool demonstrates non-exclusive licensing at massive scale, with per-unit royalties customized by application (e.g., free for internet video, capped for broadcast, royalty-bearing for paid downloads).

#### Lessons:

- Non-exclusive pooled licensing is the dominant model for codec patents.
- Royalty structures are calibrated to application category, demonstrating sophisticated stratification.
- Transparency and predictability are critical for cross-industry adoption.

### 5.2 Case Study B: Qualcomm's Licensing Model in Mobile Communications

Qualcomm historically licensed its standard-essential mobile patents non-exclusively under FRAND terms. Its licensing program produced historically high royalty rates — reaching approximately 5% of the handset sale price in certain jurisdictions — generating the bulk of its profit despite the company not manufacturing handsets. This model attracted scrutiny from regulators in the U.S., the European Union, China, South Korea, and India.

#### Lessons:

- Patent licensing can be a more profitable business than product manufacturing.
- Excessive royalty demands may invite antitrust intervention.
- FRAND commitments create legal constraints even where the underlying patents are valid.

### 5.3 Case Study C: University Technology Transfer (e.g., Stanford's Cohen-Boyer Recombinant DNA Patents)

Stanford University's licensing of the Cohen-Boyer recombinant DNA patents in the 1980s is widely regarded as a benchmark for university technology transfer. Stanford issued non-exclusive licenses to dozens of firms at modest royalties, with milestone payments and sublicensing clauses carefully crafted. The licenses generated hundreds of millions of dollars in royalties while enabling the rapid commercialization of biotechnology research.

#### Lessons:

- University licensing typically favours non-exclusivity to maximize diffusion.
- Hybrid royalty structures (lump-sum plus milestone plus running royalty) work well in university contexts.
- Transparent institutional policies encourage industry uptake.

### 5.4 Case Study D: A Fictional Indian Start-Up CS Scenario

A Noida-based AI start-up develops a novel attention-routing mechanism for transformer models. The firm files a patent and seeks to license the mechanism to multiple downstream product companies. The recommended licensing structure:

- Non-exclusive license.
- Field of use restricted to "on-device inference in consumer electronics."
- Royalty: 2.5% of net sales, with a minimum annual royalty of INR 1 crore.
- Sublicensing rights granted only with prior written consent.
- Audit rights annually.
- Cross-license option for improvements that read on the original patent.

This scenario reflects how Indian CS researchers will increasingly engage with licensing as they move from academic work to commercial deployment.

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## 6. Summary

This lecture has examined the licensing of patents along three dimensions:

1. **Types of Licenses.** The exclusive, sole, and non-exclusive license differ along the axes of number of permitted licensees, whether the licensor itself may practice, and whether the licensor may issue further licenses. Cross-licensing, compulsory licensing, patent pools, and sublicensing are derivative or related structures with distinct commercial uses.

**2. Licensing Agreements.** A well-drafted patent license contains a structure of clauses specifying grant scope, royalties, performance obligations, improvements, audit rights, warranties, term and termination, confidentiality, and dispute resolution. Care in the grant clause is particularly important.

**3. Royalty Structures.** Royalties may take the form of lump-sum, running (percentage of net sales), per-unit, milestone, or hybrid models. Rates are sensitive to exclusivity, field, geography, technology stage, market size, and standards-essentiality. Stacking, FRAND constraints, and tax considerations influence the final design.

For computer science researchers, licensing is not a peripheral commercial concern but a central intellectual property regime that shapes how algorithms, protocols, and software can be deployed, scaled, and shared across organizational and national boundaries.

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## 7. Exam Preparation Tips and Practice Questions

### 7.1 Preparation Tips

- Prepare a comparative table of exclusive, sole, and non-exclusive licenses; memorize the dimensions along which they differ.
- For each royalty type, be able to construct a numerical worked example.
- Map each clause of a licensing agreement to the commercial risk it addresses.
- Connect licensing concepts to the FRAND regime in standard-essential patents.
- Relate open-source licensing (GPL, Apache) to patent grant clauses — this is frequently asked.
- Be ready to defend or critique a chosen licensing structure for an invented start-up scenario.

### 7.2 Practice Questions

#### Short Answer Questions:

1. Differentiate between an exclusive and a sole patent license. Can a sole licensee sue an infringer independently?
2. What is a "running royalty"? State the formula and explain each variable.
3. State any three clauses commonly included in a patent licensing agreement.
4. What is the "smallest saleable patent-practicing unit," and why is it commercially relevant?
5. Differentiate licensing from assignment in patent law.
6. Explain the role of a minimum annual royalty in an exclusive patent license.
7. State the purpose of an audit clause in a patent licensing agreement.

8. What is a FRAND commitment, and how does it affect royalty levels for SEPs?
9. What is "royalty stacking," and how do patent pools mitigate it?
10. Why are improvement clauses typically included in technology licensing agreements?

### **Long Answer / Analytical Questions:**

1. Compare and contrast the three principal types of patent licenses. Illustrate your answer with a hypothetical scenario drawn from the ICT sector.
2. Examine the principal clauses of a well-drafted patent licensing agreement. Explain why each clause is important from either the licensor's or the licensee's perspective.
3. Discuss the principal royalty structures available in patent licensing. Using a numerical example, compute the royalty due under (a) a percentage model, (b) a per-unit model, and (c) a hybrid model. Comment on the commercial implications of each.
4. Analyze the role of non-exclusive licensing and patent pools in the diffusion of technology standards such as H.264 or 4G LTE. Refer to at least one contemporary case study.
5. What are standard-essential patents? How does the FRAND framework shape the licensing and royalty structure of SEPs? Critically evaluate whether FRAND prevents anticommons effects in standardization.
6. Design a complete licensing agreement framework for an Indian start-up seeking to license a novel artificial-intelligence patent to multiple downstream firms. Justify each clause and state the recommended royalty structure.
7. Discuss the interplay between open-source software licensing and patent licensing, with specific reference to the GPL's patent grant clause and the defensive termination provisions of the Apache License 2.0.

### **7.3 Application Exercise**

Imagine you are the licensing officer at a Tier-1 Indian engineering university. A computer science research group has developed a novel lightweight cryptographic protocol for IoT devices. Draft a licensing framework covering:

- Choice of license type (exclusive/sole/non-exclusive).
- Recommended royalty structure (with numerical justification).
- Field-of-use and territory scope.
- Performance milestones.
- Audit and reporting.
- Sublicensing policy.
- Term and termination conditions.

You may present your framework in a structured memorandum not exceeding 1,500 words.

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## 8. References and Further Reading

### Primary Statutes

- Patents Act, 1970 (Government of India), as amended by the Patents (Amendment) Act, 2005.
- 35 U.S.C. §§ 100–390 (United States Patent Statute).
- European Patent Convention (1973), as revised by the EPC 2000 Revision.

### Books and Treatises

- Cornish, W. R., Llewelyn, D., & Aplin, T. (2019). *Intellectual Property: Patents, Copyright, Trade Marks and Allied Rights* (8th ed.). Sweet & Maxwell.
- Narayanan, P. (2016). *Intellectual Property Law* (3rd ed.). Eastern Law House.
- Bently, L., & Sherman, B. (2014). *Intellectual Property Law* (4th ed.). Oxford University Press.
- Merges, R. P., & Nelson, P. (1990). On the Complex Economics of Patent Scope. *Columbia Law Review*, 90(4), 839–916.

### Journal Articles

- Lemley, M. A. (2002). Intellectual Property Rights and Standard-Setting Organizations. *California Law Review*, 90(6), 1889–1980.
- Shapiro, C. (2001). Navigating the Patent Thicket: Cross Licenses, Patent Pools, and Standard Setting. *Innovation Policy and the Economy*, 1, 119–150.
- Layne-Farrar, A., & Llobet, G. (2014). Payments to Infringers in Patent Settlements. *Review of Economics and Statistics*, 96(2), 308–321.

### Case Law

- *Alice Corp. v. CLS Bank International*, 573 U.S. 208 (2014).
- *Microsoft Corp. v. Motorola, Inc.*, 696 F.3d 872 (9th Cir. 2012) — concerning FRAND commitments.
- *Stanford v. Roche*, 563 U.S. 776 (2011) — concerning ownership of inventions and the Bayh-Dole framework.

**Online Resources**

- World Intellectual Property Organization (WIPO). *Patent Licensing: A Practical Guide*. Geneva: WIPO.
  - European Patent Office (EPO). *Licensing and Technology Transfer*. Munich: EPO.
  - United States Patent and Trademark Office (USPTO). *Patent Licensing* information modules.
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*End of Lecture 3 Notes.*